Candidate Seat no…………

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. Examination (Mechanical-CAD/CAM)

May 2018

ME756 / ME708.01 Finite Element Methods

Date: 03.05.2018, Thursday Time: 01:30 p.m. To 04:30 p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION- I

Q.1 Attempt the following

(a) Define basis function. Derive the linear basis functions for triangular element. Determine the shape functions N_1, N_2, and N_3 at the point P (10, 0) for the triangular element ABC if the coordinates of node points are as follows: A (15,-8), B (10, 5) and C (2, 0). **[06]**

(b) For the spring assemblage shown in the following figure 1, determine the displacements at node 2 and the forces in each spring element. Also determine the force F_3. Given: Node 3 displaces by an amount δ= 0.0254 m in the positive x direction because of the force F3 and K_1=K_2 =175.126 kN/mm. **[05]**

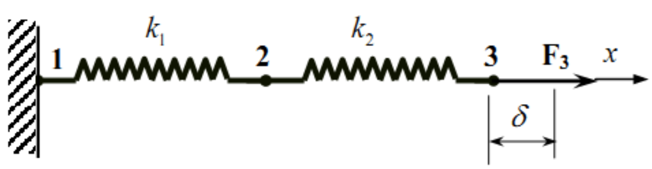


figure 1

Q.2 Attempt any one of the following **[12]**

(a) Derive strain-displacement and stress-displacement relationship for 2-D element.

For the triangular element mentioned in Q.1-(a), obtain the strain displacement relation matrix B and determine the strains ε_x, ε_y, ε_{xy}. The displacements at the nodes are

At node A: $u_x = 0.1$ $u_y = -0.050$, node B: $u_x = 0.075$ $u_y = -0.040$

and node C: $u_x = 0.080$ $u_y = -0.045$. Assume the units of Coordinates and displacement as mm. Also determine the displacement at a point P.

(b) Geology department wants to study seepage of chemically contaminated water through the ground. The governing equation of the transient phenomena is:

$$\frac{dS}{d\tau} = G_d \frac{d^2 S}{dx^2}$$

Where, S is the chemical concentration (dependent variable), τ is time, x is space variable, and G_d is known diffusivity parameter.

Develop the finite element formulation for this problem using weak form of the governing equation. Use two-noded linear element. Derive the element level characteristic matrices.

Q.3 **Attempt any one of the following** **[12]**

(a) A two member truss is as shown in Figure 2. The cross sectional area of each member is 200 mm^2 and the modulus of elasticity is 200 GPa. Determine the deflections, reactions and stress in each member.

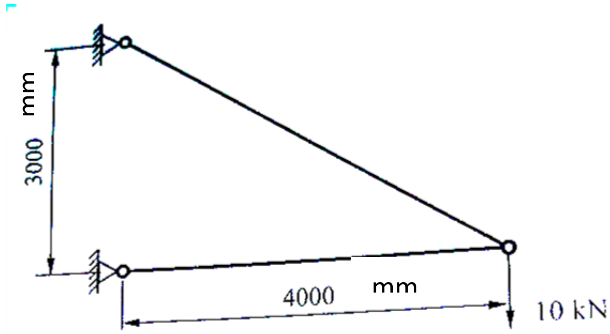
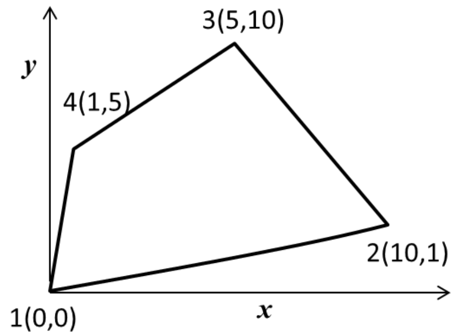


Figure 2

(b) Determine the Jacobian matrix of the given 2-dimensional quadrilateral element when transformed to the natural coordinate system. Also, determine the elements of Jacobian matrix at natural coordinates (0, 0).



SECTION- II

Q.4 **Attempt the following**

(a) Using Gaussian quadrature approach and evaluate following expression for one point (one point in each direction) and two point (two points in each direction) formulation, and compare the result with exact solution. **[06]**

$$I = \int_{-1}^{1}\int_{-1}^{1} (\xi - 1)^2 (\eta^2 - 1) d\xi d\eta .$$

(b) Explain the basic steps in Finite element analysis. **[05]**

Q.5 **Attempt any two of the following** **[12]**

(a) A steel rod of 1 cm diameter and length 6 cm with thermal conductivity K = 50 W/m^{0}C has temperature at the left end equal to 400 ^{0}C, the surrounding temperature is 30 ^{0}C, the convection heat transfer coefficient is h = 20 W/m^{0}C. The right end is insulated. Find the temperature at x = 3 cm and x = 6 cm. Use stiffness matrix.

(b) One dimensional structural element under complex loading condition is governed by the following set of equations. Find a three parameter solution using Galerkin Method.

$$\frac{d^2u}{dx^2} - 8x = 0$$

Subjected to boundary conditions: $x = 0, u = 0;\ x = 10, u = 1$

(c) Distinguish between plane stress and plane strain conditions

Q.6 **Attempt any one of the following** **[12]**

(a) A stepped metallic bar with circular cross section consist of two segments as shown in Figure 3. The modulus of elasticity of bar material is 200 GPa. Determine i) the nodal displacements ii) elements stress and iii) support reaction.

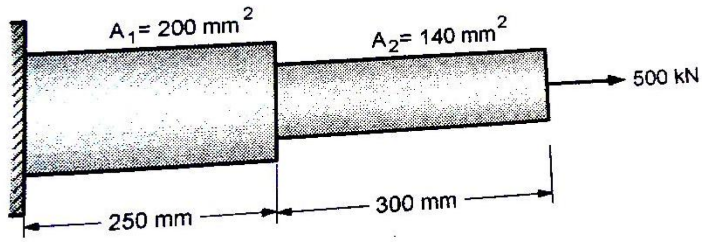


Figure 3

(b) Determine the potentials at the junctions, velocities in each sections and volumetric flow rate for the pipe shown in Figure 4. Take permeability coefficient of 1 m/sec. The potential at the end is 10 m and at the right end is 1 m.

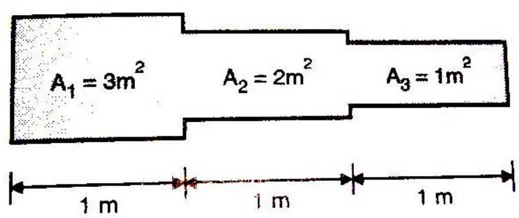


Figure 4